

FRIGID: Scaling Diffusion-Based Molecular Generation from Mass Spectra at Training and Inference Time

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Abstract

Tandem mass spectrometry is prominent in scientific discovery workflows for identifying unknown small molecules, yet high-throughput structural elucidation remains challenging. While recent autoregressive and graph diffusion models have shown promise in *de novo* elucidation, performance remains limited by poor scalability during both training and inference time. In this work, we present FRIGID, a framework with a novel diffusion language model that generates molecular structures conditioned on mass spectra via intermediate fingerprint representations and determined chemical formulae, training at the scale of hundreds of millions of unlabeled structures. We then demonstrate how forward fragmentation models enable inference-time scaling by identifying spectrum-inconsistent fragments and refining them through targeted remasking and denoising. While FRIGID already achieves strong performance with its diffusion base, inference-time scaling significantly improves its accuracy, surpassing 18% Top-1 accuracy on the challenging MassSpecGym benchmark and tripling the Top-1 accuracy of the leading methods on NPLIB1. Further empirical analyses show that FRIGID exhibits log-linear performance scaling with increasing inference-time compute, opening a promising new direction for continued improvements in *de novo* structural elucidation. FRIGID code is publicly available at <https://github.com/coleygroup/FRIGID>.

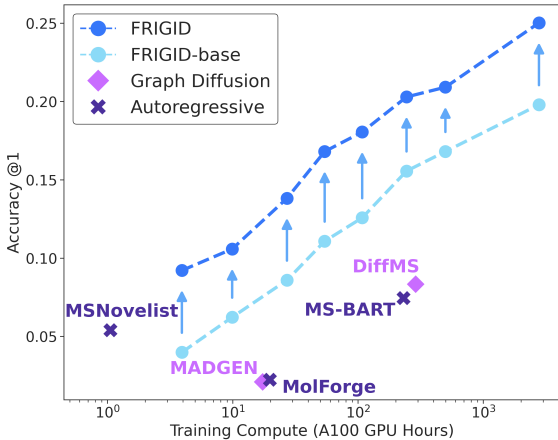


Figure 1. Top-1 exact-match accuracy as a function of training compute on the NPLIB1 dataset. Our FRIGID-base exhibits strong scaling behavior, further amplified by inference-time scaling. Data points correspond to models trained on increasingly large corpora of unlabeled molecules.

1. Introduction

Tandem mass spectrometry (MS/MS) is a cornerstone of modern analytical chemistry, enabling large-scale profiling of small molecules in human health (Gentry et al., 2024), drug discovery (Prudent et al., 2021), and environmental chemistry (Tian et al., 2021), yet determining the chemical structure from an observed spectrum remains difficult. Most deployed identification pipelines are retrieval-based, matching a query spectrum against libraries of experimental or simulated spectra, or searching a structure database using predicted molecular fingerprints. Among them, retrieval-based spectrum simulators have demonstrated strong empirical accuracy (Wang et al., 2025a) and gained wide adoption (Wang et al., 2021). However, retrieval pipelines fundamentally inherit the coverage limits of the underlying chemical libraries (e.g., PubChem) and cannot propose novel structures unless combined with an orthogonal method for molecular generation or enumeration (Qiang et al., 2026).

De novo elucidation models address this limitation by generating molecular structures directly from spectra. With the pioneering development of end-to-end models (Shrivastava et al., 2021; Butler et al., 2023; Litsa et al., 2023), two-stage

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architectures have demonstrated promising empirical performance. These methods typically consist of an encoder that first translates spectra into molecular fingerprints (Dührkop et al., 2015; Goldman et al., 2023a), followed by a decoder that translates fingerprints into molecular structures (Le et al., 2020; Stravs et al., 2022; Ucak et al., 2023; Bohde et al., 2025). Fingerprint-to-molecule decoding is a critical step with ample room for improvement. Unlike molecular generation tasks in drug discovery campaigns, the core challenge of exact reconstruction in elucidation requires strong conditioning, self-correction, and robustness to fingerprint prediction errors.

Existing fingerprint-to-molecule decoders largely repurpose architectures originally developed for *de novo* drug discovery, leaving the distinctive challenges and structure of MS/MS data insufficiently addressed. Autoregressive models (Stravs et al., 2022; Han et al., 2025; Neo et al., 2025) condition on predicted fingerprints to generate SMILES sequences, whereas fragment-based representations—such as SAFE (Noutahi et al., 2023)—remain comparatively underexplored despite their natural alignment with the fragment-centric nature of both fingerprints and MS/MS spectra. Discrete graph diffusion decoders (Bohde et al., 2025; Wang et al., 2025b) introduce stronger inductive biases, but suffer from empirically slow inference. Although fingerprint-structure pairs can be synthesized at virtually unlimited scale, enabling large-scale training as a promising path forward, systematic exploration of scaling effects remains limited to preliminary evidence in Bohde et al. (2025). Finally, existing *de novo* generators lack mechanisms for automatic correction of mistaken subgraphs, a capability that is crucial in the MS/MS context because a single token error typically leads to a wrong structural prediction.

To overcome these limitations, we introduce a diffusion language model that jointly enables scalable training, efficient inference, fragment-based generation, and inference-time error correction. As shown in Fig. 1, our architecture exhibits favorable scaling behavior with increasing training data. In addition, we propose an inference-time scaling strategy that automatically corrects erroneous substructures—a capability absent from all existing *de novo* generation methods. These design choices are motivated by the limitations of prior backbones: inference-time correction is nontrivial for standard autoregressive models, and graph diffusion models incur prohibitively slow inference (Table 2). Moving beyond naive inference-time scaling approaches such as longer denoising trajectories and stronger guidance (Karras et al., 2022; Ma et al., 2025), we leverage forward fragmentation models, including ICEBERG (Wang et al., 2025a), which attribute predicted spectral peaks to specific molecular fragments. This spectral consistency signal enables targeted masking and re-denoising of inconsistent subgraphs, rather than *post hoc* filtering of generated candidates.

Our contributions are summarized as follows:

(1) We present FRIGID, **F**ragment **R**efinement via **I**CEBERG-**G**uided **I**nference **D**iffusion, the first diffusion language model (DLM) framework for *de novo* structural elucidation (Fig. 2a). Our DLM backbone, FRIGID-base, generates 1D SAFE sequences conditioned on chemical formulae and utilizes MIST to translate spectra into fingerprints (Goldman et al., 2023a). By adopting a language-based architecture, FRIGID-base can be readily pretrained on structure-only libraries encompassing hundreds of millions of structures. To further constrain the search space by molecular formula, a unique aspect of the MS/MS context, we use NGBoost (Duan et al., 2020) to model the distribution of sequence lengths for a given molecular formula. In experiments, we demonstrate that FRIGID-base achieves strong performance even without any inference-time scaling and is significantly faster than peer diffusion models.

(2) We then demonstrate how forward fragmentation models such as ICEBERG can be leveraged for spectrum-structure cycle consistency by simulating spectra of generated molecules, identifying hallucinated fragments (i.e., mistaken subgraphs) that lack correspondence to experimental signals, and performing targeted remasking of subgraphs responsible for these inconsistencies (Fig. 2b). This enables sample-efficient inference-time scaling that prioritizes inconsistent subgraphs, rather than simply increasing the number of diffusion steps or random masking and denoising.

(3) FRIGID achieves significant performance improvements over both diffusion and autoregressive baselines on established *de novo* generation benchmarks. On the NPLIB1 data (Dührkop et al., 2021), FRIGID triples the Top-1 exact-match accuracy and achieves improved structural similarity even in cases where the true molecule is not recovered. On the challenging MassSpecGym benchmark (Bushuiev et al., 2024), FRIGID offers substantial performance improvements, including a 70% relative boost in Top-1 exact-match accuracy. This work represents foundational progress towards the long-standing goal of computational structural elucidation from mass spectra.

2. Related Work

Generative molecular design. Molecular representation is an important design factor for modern molecular generators. One family of models chooses linearized representations, which are convenient and fit soundly with infrastructures developed for natural language. SMILES (Weininger, 1988) enables the use of sequence models (Gómez-Bombarelli et al., 2018; Segler et al., 2018). Fragment-based alternatives, such as SAFE (Noutahi et al., 2023), support constrained editing and more stable generation in large chemical spaces. Efforts to approach *de novo* structural elucidation

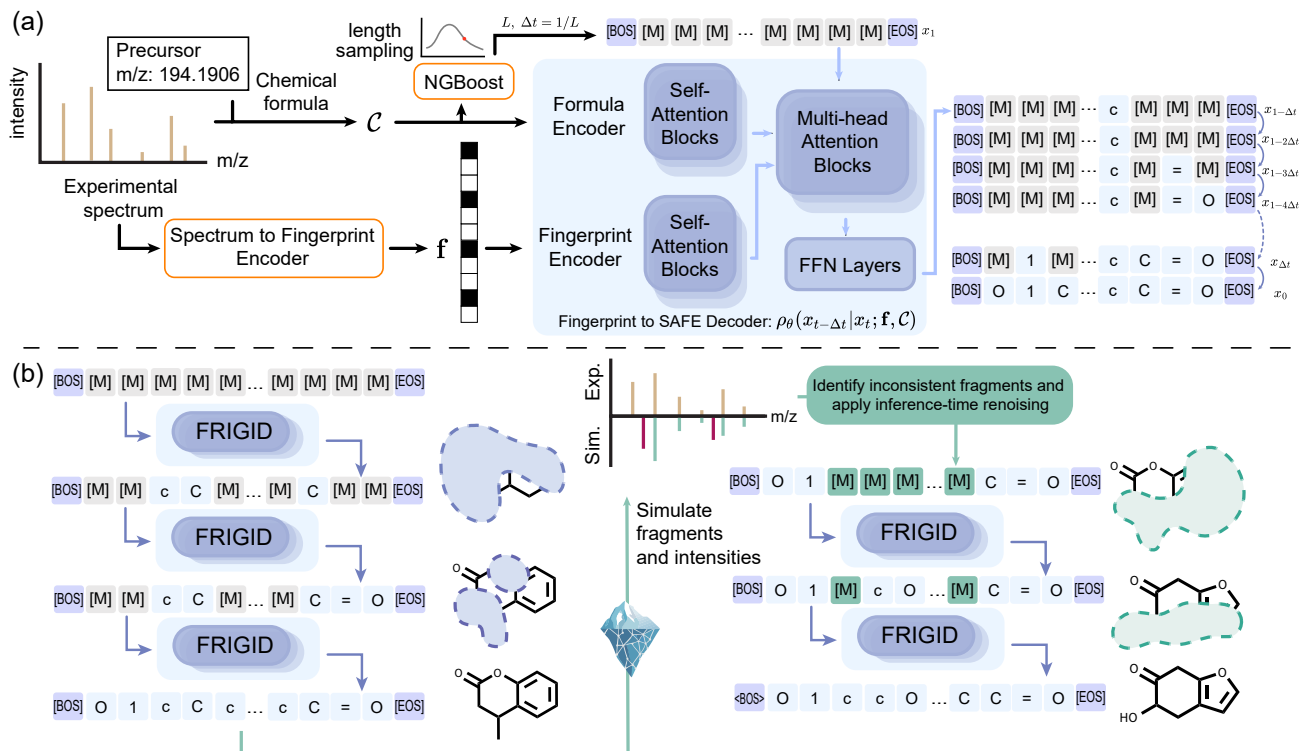


Figure 2. (a) FRIGID-base includes a masked diffusion language model (MDLM) that generates SAFE (Noutahi et al., 2023) sequences. The MDLM decoder is conditioned on a precursor formula ($\mathcal{C}$) and a MIST (Goldman et al., 2023a)-predicted molecular fingerprint ($\mathbf{f}$), injected via cross-attention. SAFE sequence lengths are predicted using NGBBoost (Duan et al., 2020). Generated structures are first filtered by chemical formula match, then ranked by Tanimoto similarity to the intermediate fingerprint $\mathbf{f}$. **(b)** The inference-time scaling pipeline enforces spectrum-structure consistency. ICEBERG model (Wang et al., 2025a) predicts simulated spectra for all generated structures. Discrepancies between the simulated and experimental spectra (hallucinations) are used to mask inconsistent tokens, which are then refined via targeted re-denoising.

using these linearized representations (Stravs et al., 2022; Litsa et al., 2023; Butler et al., 2023; Shrivastava et al., 2021) have primarily been autoregressive in nature.

Diffusion models (Sohl-Dickstein et al., 2015; Ho et al., 2020) have become a dominant paradigm for generative modeling and have been adapted to discrete and graph-structured domains (Austin et al., 2023; Vignac et al., 2023), demonstrating beneficial utility for molecular machine learning. These methods are particularly well-suited for under-specified conditional generation, where many valid outputs exist for a single condition—a common setting in chemical design. Theoretically, these models enable inference-time scaling through iterative renoising and denoising (Li et al., 2023; 2025); however, it is computationally burdensome for graph data structures compared to linearized counterparts (e.g., SMILES).

Diffusion language models (DLMs) (Li et al., 2022; Austin et al., 2023; Sahoo et al., 2024) offer a balanced trade-off in efficiency while preserving the flexibility for inference-time scaling. An exemplary DLM, GenMol (Lee et al., 2025), demonstrates strong capabilities in *unconstrained* molecu-

lar generation of SAFE sequences in the context of drug discovery. These advancements establish the foundation for the core architecture of FRIGID-base.

Structural elucidation from MS/MS. Most existing MS/MS identification pipelines remain retrieval-based. Classical methods search either experimental spectral libraries or larger structure databases using tandem MS, often by computing fragmentation trees to explain fragment relationships (Böcker & Dührkop, 2016), predicting molecular fingerprints from spectra as in CSI:FingerID (Dührkop et al., 2015), or simulating spectra from candidate structures as in CFM-ID and later neural predictors (Wang et al., 2021; Young et al., 2024; Wang et al., 2025a). These approaches are highly effective when the true molecule or a close analog is represented among the candidates, but they cannot directly recover structures absent from the reference set.

Intermediate prediction modules have therefore become central building blocks of modern elucidation pipelines. Molecular formula inference methods, including fragmentation-tree-based approaches and learned models such as MIST-CF and BUDDY (Goldman et al., 2023b; Xing et al.,

2023), help constrain the candidate search space. Fingerprint prediction from spectra, originally popularized by CSI:FingerID (Dührkop et al., 2015) and more recently advanced by transformer-based spectrum encoders such as MIST (Goldman et al., 2023a), converts MS/MS into structural descriptors that are straightforward to match against chemical databases or pass to downstream generators.

To move beyond retrieval, *de novo* generators attempt to construct molecular structures from spectra themselves. Existing spectrum-to-molecule models span end-to-end approaches, such as MassGenie, Spec2Mol, and MS2Mol, that directly map spectra to molecular strings (Shrivastava et al., 2021; Litsa et al., 2023; Butler et al., 2023), as well as the now-dominant two-stage spectra-to-fingerprint-to-structure paradigm built around learned spectrum encoders such as MIST (Goldman et al., 2023a). Representative examples include MSNovelist, which pairs CSI:FingerID with an autoregressive SMILES decoder (Stravs et al., 2022), pipelines using general fingerprint decoders such as Neuraldecipher or MolForge (Le et al., 2020; Ucak et al., 2023; Neo et al., 2025), graph-based diffusion systems such as DiffMS (Bohde et al., 2025), scaffold-guided generators such as MADGEN (Wang et al., 2025b), and unified sequence models such as MS-BART (Han et al., 2025). In parallel, forward structure-to-spectrum models such as ICEBERG provide strong *in silico* spectral prediction for candidate reranking (Wang et al., 2025a). FOAM (Manjrekar et al., 2026) further shows that forward fragmentation information can benefit inverse *de novo* generation. FRIGID combines these two lines: it adopts the modern *de novo* spectra-to-fingerprint-to-structure paradigm, while also leveraging the forward-modeling capability of ICEBERG *inside* the generative loop for targeted iterative refinement rather than only for *post hoc* scoring.

3. FRIGID-Base Diffusion Model

3.1. Diffusion language generation

As shown in Fig. 2, we cast *de novo* structural elucidation from MS/MS as a discrete conditional generation problem. Given a mass spectrum $\mathcal{S}$, FRIGID-base models a conditional distribution over SAFE strings (Noutahi et al., 2023),

$$p_{\theta}(\mathbf{x}_0 | \mathcal{S}), \quad \mathbf{x}_0 = (x_{0,1}, \dots, x_{0,L}) \in \mathcal{V}^L, \quad (1)$$

where $\mathbf{x}_0$ is a SAFE string tokenized by byte-pair encoding (BPE), decomposing the string into frequently occurring substructure fragments, and $\mathcal{V}$ is the resulting fixed vocabulary containing additional special tokens: [BOS], [EOS], [MASK], and [PAD]. In practice, we turn $\mathcal{S}$ into compact conditioning signals: a chemical formula $\mathcal{C}$ inferred from MS1 and precursor information, and a spectrum-derived fingerprint $\mathbf{f}$ predicted by MIST (Goldman et al., 2023a) (see details in Sec. 3.2). Instead of learning $p_{\theta}(\mathbf{x}_0 | \mathcal{S})$ directly,

our network learns $p_{\theta}(\mathbf{x}_0 | \mathbf{f}, \mathcal{C})$.

Masked diffusion language modeling. To parameterize $p_{\theta}(\mathbf{x}_0 | \mathbf{f}, \mathcal{C})$ non-autoregressively, we adopt a BERT-style masked diffusion language model (MDLM) following Devlin et al. (2019); Lee et al. (2025), as the denoiser over SAFE tokens. Intuitively, MDLM defines a forward corruption process that gradually converts each token into a [MASK], and learns a reverse denoiser that imputes masked positions conditioned on the unmasked context and, in our case, the spectrum-derived conditions as well.

Let $\mathbf{x}_0$ denote a clean SAFE token sequence. For a noise level $t \in [0, 1]$, the forward process produces $\mathbf{x}_t$ by independently masking tokens with probability $1 - \alpha(t)$:

$$q_t(\mathbf{x}_t | \mathbf{x}_0) = \prod_{i=1}^L q_t(x_{t,i} | x_{0,i}),$$

$$q_t(x_{t,i} | x_{0,i}) = \begin{cases} \alpha(t), & x_{t,i} = x_{0,i}, \\ 1 - \alpha(t), & x_{t,i} = [\text{MASK}], \\ 0, & \text{otherwise,} \end{cases} \quad (2)$$

where $\alpha(t) = \exp((1-t)\log\alpha_{\max} + t\log\alpha_{\min})$ is a log-linear exponential noise schedule with $\alpha_{\max} = 1$ and $\alpha_{\min} = 10^{-3}$. The reverse model is a conditional denoiser that predicts the original tokens:

$$p_{\theta}(\mathbf{x}_0 | \mathbf{x}_t, \mathbf{f}, \mathcal{C}) = \prod_{i=1}^L p_{\theta}(x_{0,i} | \mathbf{x}_t, \mathbf{f}, \mathcal{C}), \quad (3)$$

implemented as a Transformer encoder that outputs per-token categorical logits over $\mathcal{V}$. Our network has $N_{\text{layers}}=12$ Transformer blocks, hidden size $d=896$, and $H=14$ attention heads; the maximum token length is set to 256.

Training objective. Following MDLM (Sahoo et al., 2024), we train by sampling $t \sim U(0, 1)$ and minimizing the masked-token negative log-likelihood (ignoring [PAD]):

$$\mathcal{L}_{\text{MDLM}}(\theta) = \mathbb{E}_{\mathbf{x}_0 \sim \mathcal{D}, t \sim U(0,1)} \mathbb{E}_{\mathbf{x}_t \sim q_t(\cdot | \mathbf{x}_0)} \left[- \sum_{i: x_{t,i} = [\text{MASK}]} \log p_{\theta}(x_{0,i} | \mathbf{x}_t, \mathbf{f}, \mathcal{C}) \right]. \quad (4)$$

In practice, we use continuous-time training with antithetic time sampling for variance reduction and an exponential moving average (EMA) of model parameters for stability.

Inference. To sample from the model, we first predict the target length L (details for length prediction are in Sec. 3.2) and initialize $\mathbf{x}_1$ from a masked prior. For each step t , the conditional denoiser predicts logits $\mathbf{z}_{t,i} \in \mathbb{R}^{|\mathcal{V}|}$ for each i in masked positions $\mathcal{M}_t = \{i : x_{t,i} = [\text{MASK}]\}$, inducing a categorical distribution $\boldsymbol{\pi}_{t,i} = \text{softmax}(\mathbf{z}_{t,i})$ with components $\pi_{t,i,v}$ for each $v \in \mathcal{V}$. We draw provisional tokens $\hat{x}_{t,i} \sim \text{Cat}(\boldsymbol{\pi}_{t,i})$ and compute a confidence

score $p_{t,i} = \max_{v \in \mathcal{V}} \pi_{t,i,v}$ (subsequently perturbed with annealed Gumbel noise to yield $\hat{p}_{t,i}$), then unmask the single *most confident position* $i^* = \arg \max_{i \in \mathcal{M}_t} \hat{p}_{t,i}$ while keeping all other positions masked (Sahoo et al., 2024; Lee et al., 2025). After L steps, we decode the resulting SAFE sequence and deterministically convert it to SMILES, from which we generate RDKit objects for evaluation. Inference-time scaling design choices are discussed in Sec. 4.

3.2. Model architecture

Our FRIGID-base architecture couples (i) an MS/MS encoder that produces conditioning fingerprint signals and (ii) a diffusion language backbone that decodes SAFE sequences under these conditions.

Conditioning signals from MS/MS. Given a spectrum $\mathcal{S}$, we use the MIST encoder (Goldman et al., 2023a) to predict $\mathbf{f}$, a 4096-bit radius-2 Morgan fingerprint (Rogers & Hahn, 2010). The chemical formula $\mathcal{C}$ can typically be inferred upstream from high-resolution MS1 measurements using dedicated formula annotation tools (Böcker & Dührkop, 2016; Xing et al., 2023; Goldman et al., 2023b).

Conditioner embedding and cross-attention. We encode and concatenate both $\mathcal{C}$ and $\mathbf{f}$ into a single conditioning *embedding sequence* $\mathbf{Z} \triangleq [\mathbf{Z}^{(c)}, \mathbf{Z}^{(f)}]$ and inject it via cross-attention in every multi-head Transformer block. For the formula, we represent $\mathcal{C}$ as a count vector $\mathbf{y} \in \mathbb{N}^{|\mathcal{E}|}$ over a fixed element set $\mathcal{E}$ ($|\mathcal{E}|=30$ in our implementation) and embed each element type together with its count, yielding a fixed-length formula embedding sequence $\mathbf{Z}^{(c)} \in \mathbb{R}^{|\mathcal{E}| \times d}$ and a padding mask for zero-count elements.

For the fingerprint $\mathbf{f}$, we identify a set of active bit indices by applying a softmax function to the MIST output logits and selecting those exceeding a threshold of $\vartheta = 0.187^1$. We retain a maximum of 256 indices following the canonical bit order, and map each index to a learnable embedding. We then apply 3 layers of self-attention without positional encodings to these embeddings, making the network permutation-invariant while learning interactions among active substructures, yielding $\mathbf{Z}^{(f)} \in \mathbb{R}^{L_f \times d}$ with $L_f \leq 256$ and a padding mask. In our cross-attention layer, as shown in Fig. 2, we concatenate $\mathbf{Z}^{(c)}$ and $\mathbf{Z}^{(f)}$ along the conditioning-length dimension (and concatenate their padding masks accordingly), resulting in a total conditioning length $L_{\text{cond}} = |\mathcal{E}| + L_f$. Each cross-attention layer is followed by an output projection and residual layer normalization (Vaswani et al., 2017).

¹The threshold ϑ is set at 0.187 as a prior-informed calibration strategy to ensure that the overall activation density of the MIST model’s predicted fingerprints aligns with the empirical frequency of on-bits in the validation set, thereby optimizing the trade-off between precision and recall through distributional alignment without any test-set information leakage.

Token length prediction. The diffusion process requires a target length L to initialize the masked sequence $\mathbf{x}_1$. We learn a formula-to-length predictor $p_\phi(L | \mathcal{C})$ using NGBoost (Duan et al., 2020), which uses a boosted ensemble of regression trees with natural-gradient updates to predict the parameters of a conditional Normal distribution from an element-count feature vector $\mathbf{a}(\mathcal{C}) \in \mathbb{R}^{|\mathcal{E}|}$. This is well-suited as a light-weight model for the length prediction task, where L is inherently uncertain (multiple structures can share $\mathcal{C}$ but yield different tokenization lengths). At inference, we obtain $(\mu_\phi(\mathcal{C}), \sigma_\phi(\mathcal{C}))$ and sample $\tilde{L} \sim \mathcal{N}(\mu_\phi(\mathcal{C}), \lambda \sigma_\phi^2(\mathcal{C}))$, $L = \text{round}(\tilde{L})$, clipped to a valid range, and create a fully-masked $\mathbf{x}_1 = [\text{[BOS]}, [\text{MASK}], \dots, [\text{MASK}], [\text{EOS}]]$ with $L+2$ tokens (including $[\text{BOS}]/[\text{EOS}]$), before padding to the batch maximum. The scale factor λ controls the sampling variance around $\mu_\phi(\mathcal{C})$, trading off exploration of plausible lengths against wasted compute on implausible lengths.

4. Cycle-Consistent Inference-Time Scaling

While our diffusion decoder provides a strong prior over valid chemical structures, the model may still generate structures that are inconsistent with the observed spectrum. To mitigate this, we introduce an iterative refinement strategy for spectrum-structure cycle consistency (Fig. 3).

4.1. Forward fragmentation simulation

ICEBERG (Wang et al., 2025a) is a state-of-the-art forward fragmentation model that predicts tandem mass spectra from molecular structures, working in the opposite direction of FRIGID. Driving its strong empirical accuracy in spectral prediction, ICEBERG simulates spectra by predicting the most likely fragment substructures of the candidate structure and scores each fragment with a predicted intensity. Formally, an ICEBERG simulated spectrum is a set of k tuples $\mathcal{S}_{\text{sim}} = \{(\mathcal{G}_j, m_j, I_j)\}_{j=1}^k$, one tuple per peak in the simulated spectrum. For each peak j , $\mathcal{G}_j$ is a fragment (sub-graph) of the original molecule, m_j is the mass-to-charge ratio of the fragment $\mathcal{G}_j$, and I_j is the predicted intensity. Our core insight is that we can use ICEBERG’s predicted fragments to determine which atoms in a generated structure are incompatible with the observed spectrum.

4.2. ICEBERG-guided error refinement

We run inference-time scaling for R rounds. In the first round, we simply generate B samples from the FRIGID-base with different sequence lengths $L_i \sim \mathcal{N}(\mu_\phi(\mathcal{C}), \lambda \sigma_\phi^2(\mathcal{C}))$. In all future rounds, we use ICEBERG for targeted remasking and denoising.

Let $\mathcal{X}_{r-1}$ be the set of all generated molecules in rounds $1, \dots, r-1$. In each round $r > 1$, we select the top M

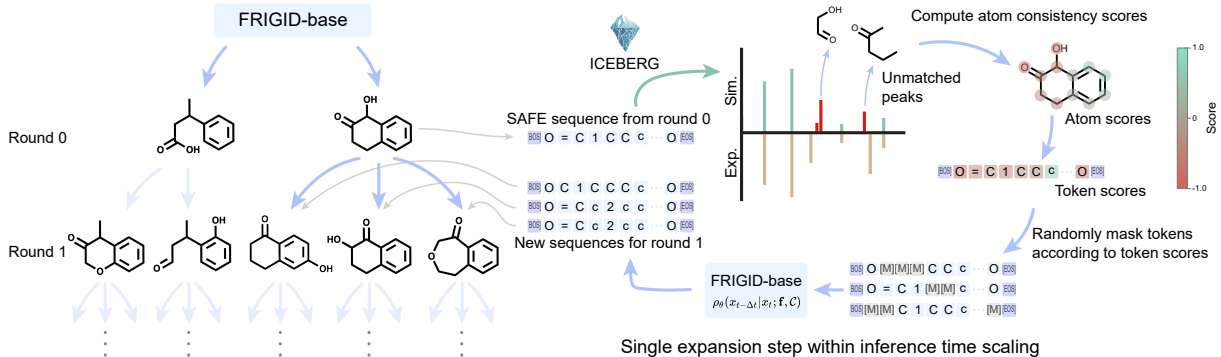


Figure 3. Mechanism of ICEBERG-guided inference-time scaling. **Left:** the process first comprises a beam search, starting with structures generated by FRIGID-base (Round 0). **Right:** at each expansion step, we use ICEBERG to simulate a mass spectrum, compare it against the experimental spectrum to identify hallucinated peaks—peaks present in simulation but missing in the experiment. Atom-wise consistency scores (S_a) are collected based on an atom’s presence in fragments corresponding to matched or hallucinated peaks (Eq. (5)). These scores are aggregated at the token-level to define a masking probability P_{mask} (Eq. (6)), to remask inconsistent substructures. The masked sequence is then re-denoised by FRIGID-base to generate refined candidates, constructing candidates for Round 1.

Algorithm 1 ICEBERG-Guided Inference-Time Scaling

Input: Observed Spectrum $\mathcal{S}_{\text{obs}}$, MIST-Predicted Fingerprint $\mathbf{f}$, Rounds R , Batch size B , Top- M count M , Number of renoised samples N

$$\mathcal{X}_{\text{pool}} \leftarrow \emptyset$$
for $r = 1$ **to** R **do****if** $r = 1$ **then**

Sample lengths $L_i \sim \mathcal{N}(\mu_\phi(\mathcal{C}), \lambda\sigma_\phi^2(\mathcal{C}))$

$\mathcal{X}_{\text{pool}} \leftarrow$ Generate B samples $\mathcal{X}_{\text{new}}$ using FRIGID-base with lengths L_i

else

$$\mathcal{X}_{\text{top}} \leftarrow \arg \max_m \underset{\mathbf{x} \in \mathcal{X}_{\text{pool}}}{\text{TaniSim}(fp(\mathbf{x}), \mathbf{f})}$$
$$\mathcal{X}_{\text{new}} \leftarrow \emptyset$$
for $\mathbf{x} \in \mathcal{X}_{\text{top}}$ **do**
$$\mathcal{S}_{\text{sim}} \leftarrow \text{ICEBERG}(\mathbf{x})$$

Compute atom consistency scores S_a using $\mathcal{S}_{\text{sim}}$ and $\mathcal{S}_{\text{obs}}$ using Eq. (5)

Calculate token masking probabilities P_{mask} from S_a using Eq. (6)

Construct N renoised inputs $\tilde{\mathbf{x}}$ by masking $\mathbf{x}$ using P_{mask}

$$\mathcal{X}_{\text{new}} \leftarrow \mathcal{X}_{\text{new}} \cup \text{Denoise}(\tilde{\mathbf{x}}|\mathbf{f})$$
end for**end if**
$$\mathcal{X}_{\text{pool}} \leftarrow \mathcal{X}_{\text{pool}} \cup \mathcal{X}_{\text{new}}$$
end for**return** $\mathcal{X}_{\text{pool}}$

molecules $\mathcal{X}_{\text{top}}$ ranked by Tanimoto similarity to the MIST-predicted fingerprint $\mathbf{f}$. For each sample $\mathbf{x}_0^{(i)} \in \mathcal{X}_{\text{top}}$, we then use ICEBERG to simulate the spectrum $\mathcal{S}_{\text{sim}}$ and attempt to match the simulated peaks in $\mathcal{S}_{\text{sim}}$ to peaks in the observed spectrum $\mathcal{S}_{\text{obs}} = \{(m_{\text{obs}}, I_{\text{obs}})\}_{i=1}^l$ within a

20 ppm (parts-per-million) tolerance as $\mathcal{P}_{\text{match}}$, and (with slight abuse of notation) we label all unmatched peaks $\mathcal{P}_{\text{halluc}} = \mathcal{S}_{\text{obs}} \setminus \mathcal{P}_{\text{match}}$ as hallucinations.

We then compute a consistency score S_a for every atom a in sample $\mathbf{x}_0^{(i)}$. Let $\text{atoms}(\mathcal{G}_j)$ be the set of atoms involved in fragment $\mathcal{G}_j$. Atoms can be (and usually are) part of multiple fragments. We define the score for each atom a by aggregating over the fragments a is in:

$$S_a = \sum_{\mathcal{G}_j \in \mathcal{S}_{\text{sim}}} \mathbb{I}[a \in \text{atoms}(\mathcal{G}_j)] \cdot \delta(\mathcal{G}_j), \quad (5)$$

where $\delta(\mathcal{G}_j)$ is a reward function that assigns +1 if $\mathcal{G}_j \in \mathcal{P}_{\text{match}}$ and $-\gamma$ if $\mathcal{G}_j \in \mathcal{P}_{\text{halluc}}$. Intuitively, atoms that frequently participate in spectral-consistent peaks accumulate positive scores while atoms in hallucinated fragments accumulate negative scores. The parameter γ allows us to tradeoff between rewarding matched atoms and penalizing spectrum-inconsistent atoms. We then normalize the atom-wise consistency scores such that $\tilde{S}_a \in [-1, 1]$.

Since our FRIGID-base operates over SAFE strings and each token may cover multiple atoms (e.g. in a ring or functional group) or similarly connect them (e.g., bond order tokens), we must aggregate atom consistency scores into token-level consistency scores. Let A_j be the set of atoms associated with token x_j . The score for token x_j is determined by the worst atom it covers, $S_{\text{token}}(j) = \min_{a \in A_j} \hat{S}_a$.

This min aggregation ensures that if any part of the token is predicted to be responsible for a hallucinated fragment, it will still receive a negative score. Finally, we convert token scores into masking probabilities $P_{\text{mask}}(j)$ for each token:

$$P_{\text{mask}}(j) = \begin{cases} p_{\text{max}} \cdot |S_{\text{token}}(j)| & \text{if } S_{\text{token}}(j) \leq 0, \\ 0 & \text{if } S_{\text{token}}(j) > 0. \end{cases} \quad (6)$$

For each $\mathbf{x}_0^{(i)} \in \mathcal{X}_{\text{top}}$, we construct N renoised inputs $\mathbf{x}_t^{(i)}$ for refinement by randomly and independently masking each token $x_j \in \mathbf{x}_0^{(i)}$ with probability $P_{\text{mask}}(j)$. These new samples are then denoised with our backbone and added to $\mathcal{X}_r$ to be ranked in the following round. The full FRIGID inference-time scaling approach is detailed in Algorithm 1.

5. Experiments

5.1. Evaluation metrics

We adopt the standard *de novo* generation metrics of Bushuiev et al. (2024) used by all recent works (Wang et al., 2025b; Bohde et al., 2025; Han et al., 2025):

- **Top- k accuracy:** measures whether the true molecule is ranked in the Top- k predictions
- **Top- k maximum Tanimoto similarity:** measures the highest structural similarity to the true molecule among the Top- k predictions
- **Top- k minimum MCES:** measures the smallest graph edit distance (maximum common edge subgraph) to the true molecule among the Top- k predictions

In Appendix A.1, we report additional close/meaningful match metrics established by Butler et al. (2023).

5.2. Datasets and baselines

Datasets and benchmarks. Following prior works, we evaluate FRIGID on the NPLIB1 (Dührkop et al., 2021) and MassSpecGym (Bushuiev et al., 2024) benchmarks. The MassSpecGym benchmark poses a challenging out-of-distribution shift with a scaffold-based test split that ensures no training molecules have an MCES < 10 compared to any test molecules. While historically some works also evaluated on NIST datasets (NIST, 2023), since this reference library is not publicly available, we focus on these two open-source libraries to enable comparison to the most recent works. In line with peer methods, we focus on the setting where the ground-truth chemical formula is provided.

FRIGID training. While prior works typically train for many epochs on a smaller dataset such as the 2.8M unlabeled molecules used by DiffMS and MIST + MolForge, we draw from LLM pretraining and train for a single epoch on a large corpus up to hundreds of millions of unlabeled structures. Specifically, we train our FRIGID-base DLM on up to 1 billion structures from Noutahi et al. (2023). We ensure that all NPLIB1 and MassSpecGym test and validation molecules are removed from the training set to ensure our model never sees even unlabeled evaluation structures. ICE-BERG models, which are used for inference-time scaling, are trained only on the training subset of each dataset. More details about model training are provided in Appendix B.2.

Peer methods. Prior works fall into two categories: Spec2Mol (Litsa et al., 2023), MIST + Neuraldecipher (Le et al., 2020; Bohde et al., 2025), MIST + MolForge (Neo et al., 2025), MIST + MSNovelist (Stravs et al., 2022), and MS-BART (Han et al., 2025) are all autoregressive models that predict SMILES/SELFIES sequences from spectra, while MADGEN (Wang et al., 2025b), DiffMS (Bohde et al., 2025), and FOAM (Manjrekar et al., 2026) are graph-based approaches, with FOAM further augmenting a forward-model-guided iterative optimization pipeline using DiffMS-generated + PubChem formula-matched seed candidates. Additionally, all baselines except for Spec2Mol and MADGEN rely on MIST (Goldman et al., 2023a) to extract dense structural features from spectra.

Because Neo et al. (2025) do not report results on NPLIB1 and do not have publicly available code, we re-implemented MIST + MolForge and report our reproduced results on both benchmarks. In Appendix A.3, we further analyze a batched-inference bug that can substantially inflate MIST + MolForge results on MassSpecGym and clarify why the corrected numbers reported in Table 1 are the appropriate comparison. Mismetti et al. (2026) recently developed a test-time tuning strategy for *de novo* structural elucidation; however, we exclude this method from comparison because it pretrains its model on a dataset (Alberts et al., 2024) without confirmation that NPLIB1 and MassSpecGym test structures are excluded during training. We provide more details on this exclusion in Appendix A.6. We also note that their test-time tuning is orthogonal to our inference-time scaling, and future works may benefit from combining both ideas.

5.3. Results and Discussion

Main results. As seen in Table 1, both FRIGID-base and FRIGID achieve leading performance across both benchmarks, achieving state-of-the-art results across 9 of the 12 metrics. FRIGID triples the Top-1 accuracy of all previous works on NPLIB1 and achieves improved structural similarity even in cases where the true molecule could not be recovered. On the more challenging MassSpecGym benchmark, FRIGID is the first to surpass 18% Top-1 accuracy. While FRIGID achieves an improvement of over $7\times$ in exact-match accuracy compared to DiffMS and FOAM, these graph-based models still maintain competitive performance on the graph-based distance metric, MCES. In Appendix A.1 and Appendix A.2–A.5, we report additional close/meaningful-match metrics, inference-time scaling ablations, robustness analyses, and direct spectra-conditioning ablations. Our DLM achieves strong performance on both benchmarks by itself; however, our proposed inference-time scaling offers significant performance improvements across all benchmarks and metrics.

Table 1. *De novo* structural elucidation performance on NPLIB1 (Dührkop et al., 2021) and MassSpecGym (Bushuiev et al., 2024) datasets, assuming known chemical formulae. The best performing model for each metric is **bold** and the second best is underlined. † indicates our implementations of baseline approaches without public code. Methods are approximately ordered by performance.

Model	Top-1			Top-10		
	Accuracy ↑	MCES ↓	Tanimoto ↑	Accuracy ↑	MCES ↓	Tanimoto ↑
NPLIB1						
Spec2Mol	0.00%	48.57	0.12	0.00%	17.58	0.16
MADGEN	2.10%	20.56	0.22	2.39%	12.69	0.27
MIST + Neuraldecipher	2.32%	10.26	0.35	6.11%	8.53	0.43
MIST + MolForge [†]	2.24%	14.16	0.40	5.11%	10.96	0.47
MIST + MSNovelist	5.40%	13.10	0.34	11.04%	8.57	0.44
MS-BART	7.45%	9.66	0.44	10.99%	8.31	0.51
DiffMS	8.34%	10.46	0.35	15.44%	8.54	0.47
FRIGID-base (ours)	<u>19.80%</u>	<u>7.88</u>	<u>0.54</u>	<u>24.41%</u>	<u>6.20</u>	<u>0.61</u>
FRIGID (ours)	25.03%	7.10	0.58	33.37%	5.56	0.64
MassSpecGym						
MIST + MSNovelist	0.00%	39.84	0.06	0.00%	18.83	0.15
Spec2Mol	0.00%	36.78	0.12	0.00%	36.02	0.16
MIST + Neuraldecipher	0.00%	22.93	0.14	0.00%	21.76	0.16
MADGEN	1.31%	27.47	0.20	1.54%	16.84	0.26
MS-BART	1.07%	16.47	0.23	1.11%	15.12	0.28
DiffMS	2.30%	13.96	0.28	4.25%	11.68	0.39
FOAM (DiffMS + PubChem)	1.50%	12.21	0.35	10.28%	6.06	0.53
MIST + MolForge [†]	10.73%	22.15	0.37	14.48%	17.88	0.41
FRIGID-base (ours)	<u>16.09%</u>	15.02	<u>0.41</u>	<u>18.19%</u>	11.95	0.46
FRIGID (ours)	18.29%	<u>13.49</u>	0.43	22.00%	<u>11.65</u>	<u>0.47</u>

Computational efficiency. Table 2 compares the inference runtime of the FRIGID backbone against other diffusion models. Despite using an architecture with 247.0M parameters, FRIGID achieves an inference speed of 6.58 seconds per spectrum, 20× faster than the graph-based DiffMS (131.2 s/spectrum) and over 3.6× faster than MADGEN (23.80 s/spectrum). This significant speedup can be attributed to the efficiency of our parallel decoding scheme compared to the computationally expensive graph operations required by existing diffusion models. This low latency provides the computational budget necessary to perform massive inference-time scaling (next paragraph) while remaining practical for large-scale spectral annotation.

Inference-time compute scaling. Fig. 4 highlights the trade-off between identification accuracy and inference cost. FRIGID exhibits a clear log-linear scaling trend (blue dashed line), where increasing the compute budget for refinement consistently boosts Top-1 accuracy. This full scaling yields a 14–26% relative improvement over FRIGID-base (Table 1). Even the base model, without inference-time correction, outperforms both autoregressive and graph diffusion baselines in accuracy while remaining orders of magnitude

Table 2. Number of parameters and inference throughput of FRIGID backbone compared with baseline diffusion models. Throughput evaluated using random samples from the NPLIB1 dataset on a single NVIDIA 6000 Ada GPU.

Model	# Parameters	Inference Latency ↓ (sec. / spectrum)	Relative Speedup
DiffMS	53.2M	131.2	1.00×
MADGEN	2.51M	23.80	5.51×
FRIGID	247.0M	6.58	20.0×

faster than graph-based methods like DiffMS. This establishes FRIGID as a flexible state-of-the-art that allows users to trade compute for precision on difficult samples.

To test whether ICEBERG-guided refinement is specific to FRIGID, we also apply the same refinement procedure to DiffMS. On NPLIB1, DiffMS improves from 8.34% to 13.33% Top-1 accuracy and from 0.35 to 0.43 Top-1 Tanimoto after 11 rounds of guidance (Table 3), confirming that the refinement mechanism itself is model-agnostic. However, this improvement requires 1423.30 seconds per sample, compared to 86.49 seconds for FRIGID, and we observe a similar qualitative trend on MassSpecGym. This reinforces

Table 3. ICEBERG-guided refinement applied to DiffMS on the NPLIB1 benchmark. ICEBERG improves DiffMS substantially, confirming that the refinement strategy is model-agnostic; however, the resulting latency is much higher than for FRIGID.

Model	Top-1			Top-10			Inference Compute (sec./sample) ↓
	Accuracy ↑	MCES ↓	Tanimoto ↑	Accuracy ↑	MCES ↓	Tanimoto ↑	
DiffMS	8.34%	10.46	0.35	15.44%	8.54	0.47	131.21
DiffMS + ICEBERG	13.33%	9.30	0.43	23.29%	6.91	0.54	1423.30
FRIGID-base (ours)	19.80%	7.88	0.54	24.41%	6.20	0.61	6.58
FRIGID (ours)	25.03%	7.10	0.58	33.37%	5.56	0.64	86.49

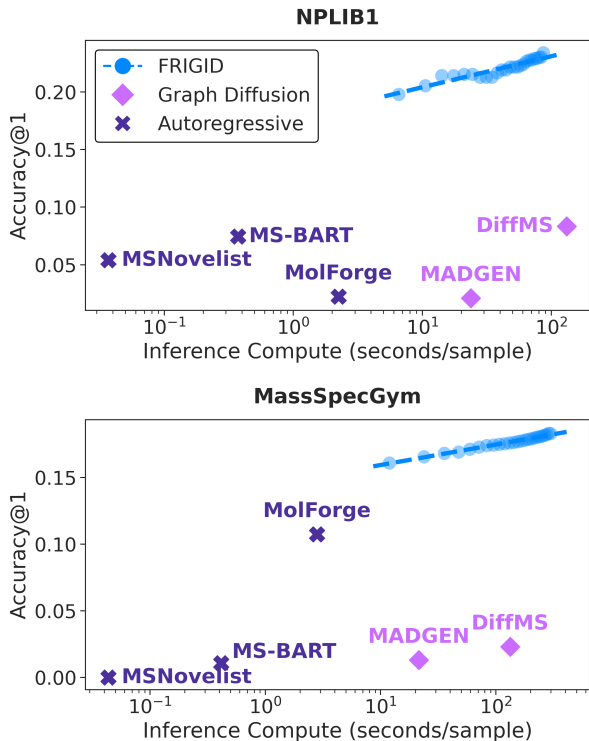


Figure 4. Inference-time scaling on NPLIB1 and MassSpecGym. The plots display Top-1 identification accuracy as a function of inference compute time for FRIGID and baseline methods. The blue dashed line illustrates FRIGID’s performance across increasing rounds of refinement, demonstrating a log-linear scaling behavior where additional compute yields higher accuracy. FRIGID significantly outperforms autoregressive baselines (e.g., MS-BART, MSNovelist) and graph diffusion models (e.g., DiffMS, MADGEN) in both accuracy and efficiency.

our core claim: forward fragmentation guidance transfers across backbones, but a sequence-based diffusion decoder is what makes iterative refinement practical. Appendix A.2 further compares ICEBERG-guided refinement to compute-matched naive scaling baselines and reports sensitivity analyses over refinement hyperparameters.

Pretraining scaling analysis. As illustrated in Figure 1, our DLM backbone exhibits strong scaling behavior with respect to training compute. We observe a log-linear relationship between the amount of compute used during pre-

training (approximated by training duration on increasingly large datasets) and the resulting Top-1 accuracy on NPLIB1. Notably, even at lower compute budgets, FRIGID pushes the accuracy significantly beyond existing baselines, achieving higher accuracy with orders of magnitude less training cost than graph-based diffusion models like MADGEN. This trend suggests that our sequence-based architecture effectively leverages large-scale unlabeled data, and that further scaling of the pretraining corpus and model size is a promising avenue for continued performance gains.

6. Conclusion

In this work, we present FRIGID, a DLM-based framework for *de novo* structural elucidation from tandem mass spectra. In favor of a sequence-based diffusion backbone, we achieve a 20× inference speedup over state-of-the-art graph diffusion methods, enabling scalable decoder pretraining on hundreds of millions of molecules without spectrum labels. Our efficient DLM architecture enables novel inference-time scaling, leveraging a neural spectral simulator to iteratively detect and correct spectral hallucinations through targeted masking and denoising. Empirically, FRIGID establishes a new state-of-the-art on the NPLIB1 and MassSpecGym benchmarks, demonstrating a log-linear relationship between inference compute and identification accuracy. These results suggest a promising new direction for continued improvements in *de novo* structural elucidation.

Software and Data

FRIGID code and models are available at <https://github.com/coleygroup/FRIGID>.

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Impact Statement

Advances in computational methods for *de novo* structure elucidation can substantially improve the identification of previously unknown compounds, including metabolites that serve as diagnostic biomarkers or illuminate underlying biological processes. These capabilities have broad potential for positive societal impact across healthcare and life sciences, with minimal foreseeable risk.

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A. Additional Experimental Results

A.1. Additional metrics

Following (Bohde et al., 2025) we report additional metrics to further contextualize how close FRIGID predictions are to the true molecule even in cases where the true structure cannot be recovered. In Table 4, we report the percentage of FRIGID samples that correspond to valid molecules as well as the percentage of Top-1 and Top-10 predictions that are “close” and “meaningful” structural matches according to the definitions of Butler et al. (2023). Specifically, a molecule is a “meaningful” match if it has an RDKit fingerprint Tanimoto similarity ≥ 0.4 compared to the reference structure, and it is a close match if the similarity is ≥ 0.675 . These thresholds were selected based on domain-expert annotations, and close structural matches can be considered to be useful in practical settings. As shown in Table 4, FRIGID achieves strong performance, including achieving close Top-1 predictions on more than half of NPLIB1 test spectra. We also note that FRIGID-base achieves similarly strong Top-10 performance and is able to consistently generate meaningful and close matches without any inference-time scaling, however inference-time scaling helps to refine close matches into exact structural matches.

Table 4. Additional evaluation of molecule validity, and percentage above domain-expert-defined Tanimoto thresholds on NPLIB1 (Dührkop et al., 2021) and MassSpecGym (Bushuiev et al., 2024) *de novo* generation datasets. The best performing model for each metric is **bold** and the second best is underlined. † indicates our implementations of baseline approaches without public code. Methods are approximately ordered by performance. Definitions of meaningful match (Tanimoto similarity of RDKit fingerprints ≥ 0.4) and close match (Tanimoto similarity of RDKit fingerprints ≥ 0.675) are taken from Butler et al. (2023).

Model	Overall	Top-1		Top-10	
	% Valid †	% Meaningful match †	% Close match †	% Meaningful match †	% Close match †
NPLIB1					
Spec2Mol	66.5%	4.09%	0.00%	8.05%	0.00%
MIST + Neuraldecipher	81.78%	65.81%	28.57%	78.14%	39.68%
MIST + MSNovelist	<u>98.60%</u>	13.72%	3.39%	53.03%	7.35%
MIST + MolForge†	100.0%	63.14%	31.76%	73.47%	43.34%
DiffMS	100.0%	77.10%	42.82%	87.50%	59.03%
FRIGID-base (ours)	100.0%	<u>79.70%</u>	<u>52.80%</u>	88.42%	<u>67.00%</u>
FRIGID (ours)	100.0%	80.20%	57.41%	<u>88.04%</u>	67.25%
MassSpecGym					
Spec2Mol	68.5%	4.09%	0%	27.70%	0%
MIST + Neuraldecipher	91.11%	53.75%	2.50%	63.37%	4.27%
MIST + MSNovelist	<u>98.58%</u>	57.17%	30.92%	70.06%	42.58%
MIST + MolForge†	100.0%	52.68%	24.58%	68.66%	29.02%
DiffMS	100.0%	78.56%	28.28%	96.67%	<u>51.74%</u>
FOAM (DiffMS+PubChem)	100.0%	<u>78.00%</u>	35.55%	<u>93.66%</u>	62.94%
FRIGID-base (ours)	100.0%	72.66%	32.86%	85.87%	40.11%
FRIGID (ours)	100.0%	75.61%	<u>34.63%</u>	86.34%	41.86%

A.2. Inference-Time Scaling Ablations

This section provides additional evidence that the gains of FRIGID arise from targeted refinement rather than simply spending more inference-time compute. We first show that ICEBERG-guided refinement transfers to a graph-diffusion backbone, then compare against compute-matched naive scaling baselines, and finally evaluate sensitivity to the main refinement hyperparameters.

A.2.1. TARGETED ICEBERG REFINEMENT OUTPERFORMS NAIVE COMPUTE SCALING

We next compare FRIGID against two compute-matched baseline scaling strategies on NPLIB1. *Random masking* preserves the same multi-round structure as FRIGID but masks the selected candidates uniformly at random instead of using ICEBERG scores. *No masking* never refines existing candidates and instead spends the full budget on repeated from-scratch generation. Both baselines improve over FRIGID-base, showing that additional inference compute is indeed useful; however, both underperform ICEBERG-guided refinement. This indicates that the benefit of FRIGID is not merely increased compute, but

rather how that compute is allocated to spectrum-inconsistent substructures.

Table 5. Compute-matched baselines on NPLIB1. Naively increasing inference-time compute improves over FRIGID-base, but targeted ICEBERG-guided refinement performs best under the same overall candidate budget.

Model	Top-1			Top-10		
	Accuracy $\uparrow$	MCES $\downarrow$	Tanimoto $\uparrow$	Accuracy $\uparrow$	MCES $\downarrow$	Tanimoto $\uparrow$
Random Masking	21.54%	7.76	0.54	26.65%	6.22	0.599
No Masking	21.17%	7.45	0.546	27.27%	5.92	0.608
FRIGID (ours)	25.03%	7.10	0.58	33.37%	5.56	0.643

A.2.2. SENSITIVITY TO REFINEMENT HYPERPARAMETERS

We evaluate sensitivity to the main refinement hyperparameters on NPLIB1. Specifically, we vary the maximum masking probability $p_{\max}$, the number of candidates refined per round (K), the number of renoised variants per candidate (M), and the hallucination penalty weight γ . Across the tested settings, performance varies only modestly, suggesting that the refinement procedure is reasonably robust and does not depend on a narrow hyperparameter regime.

Table 6. Sensitivity analysis of FRIGID refinement hyperparameters on NPLIB1. The procedure is fairly stable across nearby settings of $p_{\max}$, K , M , and γ .

γ	$p_{\max}$	K	M	Acc@1 $\uparrow$	Tanimoto@1 $\uparrow$	Acc@10 $\uparrow$	Tanimoto@10 $\uparrow$
1	0.75	12	8	25.03%	0.58	33.37%	0.64
1	0.50	12	8	22.67%	0.57	30.88%	0.64
1	0.50	6	6	23.54%	0.57	33.37%	0.65
1	0.75	16	4	24.03%	0.57	33.75%	0.65
1	0.25	12	8	24.16%	0.57	32.00%	0.64
1	1.00	12	8	23.29%	0.57	31.88%	0.64
1	0.75	8	12	24.16%	0.57	32.05%	0.64
5	0.75	12	8	23.91%	0.57	33.75%	0.64
10	0.75	12	8	24.15%	0.57	32.75%	0.62

A.3. Baseline Reassessment: MIST + MolForge

We investigated the discrepancy between our reproduced MIST + MolForge results and the much stronger MassSpecGym numbers reported in Neo et al. (2025). We identified a bug in batched MIST inference inherited from older implementations: the attention mask adds zero, rather than negative infinity, to padded positions. As a result, shorter spectra can attend to padding and collapse toward the longest spectrum in the batch. Because MassSpecGym groups spectra from the same molecule, this leakage can substantially inflate fingerprint quality and downstream reconstruction accuracy when the batch size is significantly greater than 1. We successfully reproduced the inflated behavior and confirmed that the corrected implementation yields the lower, more reliable numbers reported in Table 1. Since Neo et al. (2025) did not publicly release their MIST + MolForge implementation, we developed our own reimplementations, and release both the faulty and corrected reimplementations at <https://github.com/harrylaucngd/MIST-MolForge>.

Table 7. MassSpecGym results for MIST + MolForge under corrected and inflated batched MIST inference. The inflated setting reproduces the unexpectedly strong numbers caused by the masking bug, whereas the corrected implementation remains below FRIGID.

Model	MIST Tanimoto $\uparrow$	Top-1 Acc $\uparrow$	Top-1 Tan $\uparrow$	Top-10 Acc $\uparrow$	Top-10 Tan $\uparrow$
MIST + MolForge (corrected)	0.457	10.73%	0.37	14.48%	0.41
MIST + MolForge (inflated, batch size= 24)	0.812	31.75%	0.68	40.55%	0.74
FRIGID (ours)	0.457	18.29%	0.43	22.00%	0.47

A.4. Robustness to Imperfect Intermediate Predictions

The main paper evaluates the standard setting where the chemical formula is assumed known and the molecular fingerprint is predicted from spectra. Here we test robustness when these intermediate signals are imperfect.

A.4.1. ROBUSTNESS TO FORMULA UNCERTAINTY

To measure robustness to formula annotation errors, we use MIST-CF (Goldman et al., 2023b) to generate the top 5 formula hypotheses for each test entry. To preserve a fair compute budget, we keep the total number of generated molecules fixed and distribute that budget across the 5 candidate formulae. While performance degrades without oracle formulae, FRIGID remains reasonably robust on both NPLIB1 and MassSpecGym.

Table 8. Robustness to unknown chemical formulae. We replace the ground-truth formula with the top 5 MIST-CF hypotheses while keeping the total generation budget fixed.

Model	Top-1			Top-10		
	Accuracy $\uparrow$	MCES $\downarrow$	Tanimoto $\uparrow$	Accuracy $\uparrow$	MCES $\downarrow$	Tanimoto $\uparrow$
NPLIB1						
FRIGID-base (known formula)	19.80%	7.88	0.54	24.41%	6.20	0.61
FRIGID-base (formula unknown)	18.68%	8.78	0.52	22.91%	6.70	0.58
FRIGID (known formula)	25.03%	7.10	0.58	33.37%	5.56	0.64
FRIGID (formula unknown)	21.79%	7.94	0.55	30.14%	6.23	0.62
MassSpecGym						
FRIGID-base (known formula)	16.09%	15.02	0.41	18.19%	11.95	0.46
FRIGID-base (formula unknown)	11.60%	20.19	0.35	13.13%	14.81	0.42
FRIGID (known formula)	18.29%	13.49	0.43	22.00%	11.65	0.47
FRIGID (formula unknown)	14.28%	18.30	0.37	17.43%	14.85	0.43

A.4.2. FINGERPRINT QUALITY DISTRIBUTIONAL ANALYSIS

We also retrospectively bin test spectra by the Tanimoto similarity between the MIST-predicted fingerprint and the ground-truth fingerprint. As expected, reconstruction accuracy and structural similarity improve monotonically with fingerprint quality. Nonetheless, even when the predicted fingerprint has Tanimoto similarity below 0.5, the true molecule can still occasionally be recovered, indicating that FRIGID can tolerate nontrivial upstream fingerprint error.

Table 9. Distributional analysis of FRIGID performance as a function of upstream fingerprint quality. Test spectra are binned by the Tanimoto similarity between the MIST-predicted fingerprint and the ground-truth fingerprint.

Metric	[0.0, 0.25)	[0.25, 0.5)	[0.5, 0.75)	[0.75, 1.0]
NPLIB1				
Acc@1 $\uparrow$	0.0%	6.09%	23.55%	60.00%
Tanimoto@1 $\uparrow$	0.18	0.36	0.67	0.89
Acc@10 $\uparrow$	4.80%	13.71%	32.97%	70.24%
Tanimoto@10 $\uparrow$	0.25	0.45	0.75	0.93
MassSpecGym				
Acc@1 $\uparrow$	0.2%	3.00%	7.68%	42.44%
Tanimoto@1 $\uparrow$	0.17	0.31	0.50	0.79
Acc@10 $\uparrow$	0.3%	4.93%	11.65%	48.73%
Tanimoto@10 $\uparrow$	0.20	0.35	0.56	0.82

A.5. Direct Spectra Conditioning Ablation

The two-stage spectra-to-fingerprint-to-structure pipeline is the standard design in modern *de novo* elucidation systems, but one might ask whether direct end-to-end spectra conditioning is preferable with a stronger decoder. To test this, we fine-tune our model directly on spectra-conditioned generation and compare against the fingerprint-conditioned backbone. Direct end-to-end conditioning consistently hurts exact reconstruction, with the gap being especially large on the more challenging MassSpecGym benchmark. These results support the use of the intermediate fingerprint as a useful structural prior rather than an unnecessary decomposition.

Table 10. Direct spectra conditioning versus the standard fingerprint-conditioned pipeline. End-to-end conditioning underperforms the two-stage formulation on both benchmarks, especially on MassSpecGym.

Model	Top-1			Top-10		
	Accuracy $\uparrow$	MCES $\downarrow$	Tanimoto $\uparrow$	Accuracy $\uparrow$	MCES $\downarrow$	Tanimoto $\uparrow$
NPLIB1						
E2E FT-Base	18.30%	8.39	0.50	22.17%	6.58	0.57
E2E FT	24.28%	7.10	0.56	33.13%	5.54	0.63
FRIGID-base (fingerprint-conditioned)	19.80%	7.88	0.54	24.41%	6.20	0.61
FRIGID (fingerprint-conditioned)	25.03%	7.10	0.58	33.37%	5.56	0.64
MassSpecGym						
E2E FT-Base	6.05%	13.97	0.33	6.54%	12.33	0.39
E2E FT	11.13%	12.37	0.41	12.11%	11.18	0.44
FRIGID-base (fingerprint-conditioned)	16.09%	15.02	0.41	18.19%	11.95	0.46
FRIGID (fingerprint-conditioned)	18.29%	13.49	0.43	22.00%	11.65	0.47

A.6. Multimodal Spectroscopic Dataset Overlap

Mismetti et al. (2026) have recently developed a test-time tuning strategy that dynamically adapts a pretrained *de novo* elucidation model to the most relevant experimental spectra at inference time. We have omitted a comparison with their approach because their base model is pretrained on a dataset of simulated spectra (Alberts et al., 2024) which we found contains substantial amounts of both NPLIB1 and MassSpecGym test structures. The full results of this analysis are shown in Table 11, where we find more than 10% of both the NPLIB1 and MassSpecGym test sets overlap with the training set. Additionally, the models used to build this simulated spectra dataset were likely trained on datasets which contain true NPLIB1 and MassSpecGym test structures. Because of this, overlap, it is not possible to fairly compare the method in Mismetti et al. (2026) to FRIGID and other baselines. While we did not have the capacity to retrain their model from scratch without on cleaned data, we note that their test-time tuning strategy is orthogonal to our inference-time scaling and that future works may benefit from combining both ideas.

Table 11. Analysis of structures in the Multimodal Spectroscopic Dataset (Alberts et al., 2024) also present in the NPLIB1 and MassSpecGym test splits. Two structures are considered to be a match if they have the same connectivity-based (2D) InChIKeys.

Dataset	Train-Test Overlap	
	# of Structures	% of Test Set
NPLIB1	122	14.90%
MassSpecGym	1820	10.37%

B. Experimental Details

B.1. Model Details

As detailed in Sec. 3.2, we parametrize $p_{\theta}(\mathbf{x}_0 \mid \mathbf{f}, \mathcal{C})$ using a BERT-style masked diffusion language model (MDLM) with an additional cross-attention mechanism added in each transformer layer to inject information from the conditioning sequence. We set the MDLM backbone to a 12-layer Transformer encoder with hidden size $d=896$, $H=14$ attention heads, and feed-forward dimension $d_{\text{ff}}=3072$, using dropout 0.1 throughout. We tokenize SAFE strings using a byte-pair encoding (BPE) tokenizer; the resulting vocabulary size is $|\mathcal{V}|=1880$, and we use a maximum token length of 256 (including $[\text{BOS}]/[\text{EOS}]$), padding shorter sequences with $[\text{PAD}]$.

Formula conditioning. We encode the precursor formula $\mathcal{C}$ as a fixed-dimensional count vector over a set of 30 common elements, without normalization. For cross-attention conditioning, we represent $\mathcal{C}$ as a fixed-length embedding sequence of length 30 by summing (i) a learnable element embedding, (ii) a learnable count embedding (with counts clipped to a maximum of 200), and (iii) a learned positional embedding. We apply a conditioning mask that zeros out elements with count 0. We do not apply formula conditional dropout during training ($p_{\text{drop}, \mathbf{f}}=0$).

Fingerprint conditioning. We condition on a 4096-bit radius-2 Morgan fingerprint. We embed the fingerprint as an unordered set of active bits, retaining up to 256 indices following canonical bit order; in our training configuration, the

activation threshold is 0 (i.e., all non-zero bits are treated as active). Each active index is mapped to a learnable embedding, and we apply 3 layers of permutation-equivariant self-attention (without positional encodings) to model interactions among active substructures, yielding a fingerprint embedding sequence and a corresponding padding mask. For robustness, we apply fingerprint conditional dropout during training with probability $p_{\text{drop},C}=0.25$.

Cross-attention injection. We inject formula and fingerprint conditions via cross-attention sublayers in every Transformer block (with the same hidden size and number of heads as the backbone). We use shared cross-attention pathway for the two modalities.

B.2. Training Details

We pretrain the FRIGID-base DLM on the SAFE-GPT molecular corpus from Noutahi et al. (2023) and mask out all NPLIB1 and MassSpecGym test and validation structures using an exclusion list of connectivity-based (2D) InChIKeys. Prior studies have shown that the SAFE molecular representation can be beneficial for de novo molecular generation (Noutahi et al., 2023; Lee et al., 2025), which motivates our adoption of SAFE during pretraining. That said, our work does not perform an independent head-to-head ablation of molecular representation choice, and therefore does not directly benchmark SAFE’s claimed advantages over SMILES for molecular generation. We adopted the datasets used in these two prior works without modification, including approximately 17% of the SAFE strings in the SAFE-GPT corpus that could not be parsed into valid molecular structures. They were still retained for MDLM pretraining; because no valid molecular formula or fingerprint could be computed for these samples, their cross-attention conditioning falls back to an all-zero mask, effectively acting as a form of dropout.

We train using AdamW (Loshchilov & Hutter, 2019) with a cosine annealing schedule (Loshchilov & Hutter, 2017). Our largest model is trained on over 1 billion structures; we found that fine-tuning on NPLIB1 and MassSpecGym did not improve end-to-end performance (Table 10) and thus omit any finetuning.

In our largest set, the training configuration is as follows: we train for 400,000 optimization steps with global batch size 576. We use mixed precision (bfloat16) and clip gradients to max norm 1.0. We train in continuous time ($T=0$) with log-linear noise and antithetic time sampling, using sampling $\epsilon=10^{-3}$. We maintain an EMA of parameters with decay 0.9999 for evaluation. AdamW hyperparameters are $\beta_1=0.9$, $\beta_2=0.999$, $\epsilon=10^{-8}$, and weight decay 0, with peak learning rate 3×10^{-4} . The learning rate is linearly warmed up for 50,000 steps and then annealed with cosine decay (CosineAnnealingLR) with $T_{\text{max}}=400,000$ and $\eta_{\text{min}}=10^{-8}$. We optimize the MDLM reconstruction objective with global-mean loss aggregation, and do not use bracket-SAFE tokens in this configuration.

We do not use an auxiliary formula-matching loss in this configuration (formula loss weight 0.0), relying instead on formula filtering at inference time and the strong conditional signal provided by cross-attention.

B.2.1. ICEBERG MODEL TRAINING DETAILS

To use a forward spectrum simulation method as part of our inference-time strategy, we adopt the deep learning model, ICEBERG (Wang et al., 2025a), which, briefly, fragments candidate structures to predict the most likely fragments to appear, and scores intensities for each of these fragments to compose a simulated MS/MS spectrum. For the MassSpecGym and NPLIB1 benchmarks, we first train a version of ICEBERG specific to each dataset. Each model is only trained on the same training structures and spectra specified by each benchmark to avoid leakage of test structures at the correction stage. We include training example counts below (Table 12).

Table 12. Number of training examples used to train each version of ICEBERG.

Dataset	Training structures	Training spectra
NPLIB1	6,810	9,272
MassSpecGym	13,543	101,573

ICEBERG is a two-part model; the first model learns to predict likely fragments through bond-breaking of the candidate structure. As experimental fragments are difficult to characterize in MS/MS, this first model is therefore supervised on rule-based enumeration and annotation of fragments as collected from the MAGMA algorithm (Ridder et al., 2014). The second model learns to take the fragments, as well as the original structure, to predict intensities. Both models are trained with cross-entropy loss. To boost differentiation and ranking of candidate structures, the intensity model is also further

fine-tuned with a contrastive learning loss against decoy structures.

B.3. Inference-time Scaling Details

Our inference-time scaling procedure implements the cycle-consistent refinement pipeline described in Sec. 4. For each test spectrum, we obtain a binary fingerprint by thresholding the spectrum-predicted fingerprint probabilities with $\vartheta = 0.187$, and treat the molecular formula as known. After each refinement round, we rank the pool of unique candidates by fingerprint Tanimoto similarity to the predicted fingerprint, while prioritizing candidates whose molecular formula matches the target formula; we then report Top- k metrics on the ranked list. Refinement is run for R rounds with a per-round sample budget B . In round 1, we generate B candidates from scratch using the conditional DLM. In subsequent rounds, we refine the top K unique candidates from the current pool by constructing M renoised variants per candidate (via masking) and denoising them, yielding up to $M \cdot K$ refined samples; the remaining budget, $B - M \cdot K$, is filled with additional from-scratch generations to preserve diversity. By default, we use $B=128$, $K=12$, and $M=8$. For inference-time scaling, we sweep the number of rounds R up to 25 and evaluate intermediate rounds.

We simulate spectra using ICEBERG with collision energies $\{10, 20, 30, 40, 50, 60, 70, 80, 90\}$ (eV) and adduct $[M+H]^+$. We simulate over a range of collision energies to also cover cases where the collision energy was not recorded, selecting the collision energy that minimizes the discrepancy. We match simulated and experimental peaks within a 20 ppm tolerance and ignore the precursor peak. For each candidate, we identify a small set of informative simulated peaks (by intensity thresholding and selecting the top few peaks) and mark each peak as *matched* if it lies within the ppm tolerance of any experimental peak; otherwise it is treated as a hallucination. ICEBERG associates each simulated peak with a fragment, enabling attribution of peaks to atoms in the candidate molecule. As described in Sec. 4, atom-wise consistency scores are computed, then normalized and aggregated into token-wise consistency scores, yielded from contribution and penalty from matched and hallucinated peaks, respectively. Finally, tokens with non-positive scores are independently masked with probability $p_{\max} \cdot |S_{\text{token}}|$, where $p_{\max}=0.75$. The resulting partially masked sequence is then re-denoised under the original conditions to propose refined candidates.

We maintain a persistent candidate pool across rounds and deduplicate candidates by connectivity (stereochemistry-agnostic) identifiers to ensure refinement invests compute in genuinely new structures rather than repeated samples. To amortize cost, ICEBERG calls are batched across molecules and collision energies, and simulated spectra are cached for reuse across rounds.

B.4. Compute Benchmarking Details

For our inference compute benchmarks (in Sec. 5.3), we measure the latency in seconds/spectra on a single NVIDIA 6000 Ada GPU after an initial warmup period. Diffusion models, including FRIGID, DiffMS, and MADGEN, generate many samples and then apply some ranking produce to determine the final ranked predictions. Specifically, FRIGID generates 128 samples, DiffMS generates 100 samples, and MADGEN generates 50 samples. The reported seconds/spectra refers to the total time taken to generate all samples (which is different for each model). We omit speed comparisons to Spec2Mol (Litsa et al., 2023) and MIST + Neuraldecipher (Bohde et al., 2025; Le et al., 2020) as the inference code reproduced in Bohde et al. (2025) is not publicly available. Similarly, to benchmark the GPU hours required to train each baseline in Sec. 5.3, we train each model for one epoch on a single NVIDIA A100 GPU and multiply by the number of epochs used to train each model.

B.5. Train Compute Scaling Details

In Table 13 we provide settings and additional experimental results for each of the models in our training compute scaling study. We train each model for a single epoch with increasingly large numbers of training structures ranging from 3.2M to over 1B. For models with less than 10M training structures, we find that smaller models achieve better empirical performance. We hypothesize that larger models might be appropriate when scaling training to billions of structures.

Table 13. Compute and accuracy evaluations from progressive scaling of the pretraining set. While holding parameter count constant for the larger half of pretraining sets, Top-1 and Top-10 reconstruction and structural similarity continues to improve with additional molecules observed.

# Training Structures	# Params	Training GPU Hours	Top-1		Top-10	
			Accuracy $\uparrow$	Tanimoto $\uparrow$	Accuracy $\uparrow$	Tanimoto $\uparrow$
3.2M	82.5M	3.92	9.22%	0.39	12.58%	0.46
8M	82.5M	9.89	10.58%	0.43	14.57%	0.49
12.5M	182.2M	27.06	13.82%	0.47	18.93%	0.53
25.1M	182.2M	54.12	16.81%	0.50	22.42%	0.56
49.9M	182.2M	107.83	18.06%	0.50	22.79%	0.56
112.8M	182.2M	243.65	20.30%	0.53	26.77%	0.60
230.4M	182.2M	497.67	20.92%	0.55	28.02%	0.61
1.0B	247.0M	2777.77	25.03%	0.58	33.37%	0.64

C. FRIGID Predictions

C.1. NPLIB1 Predictions

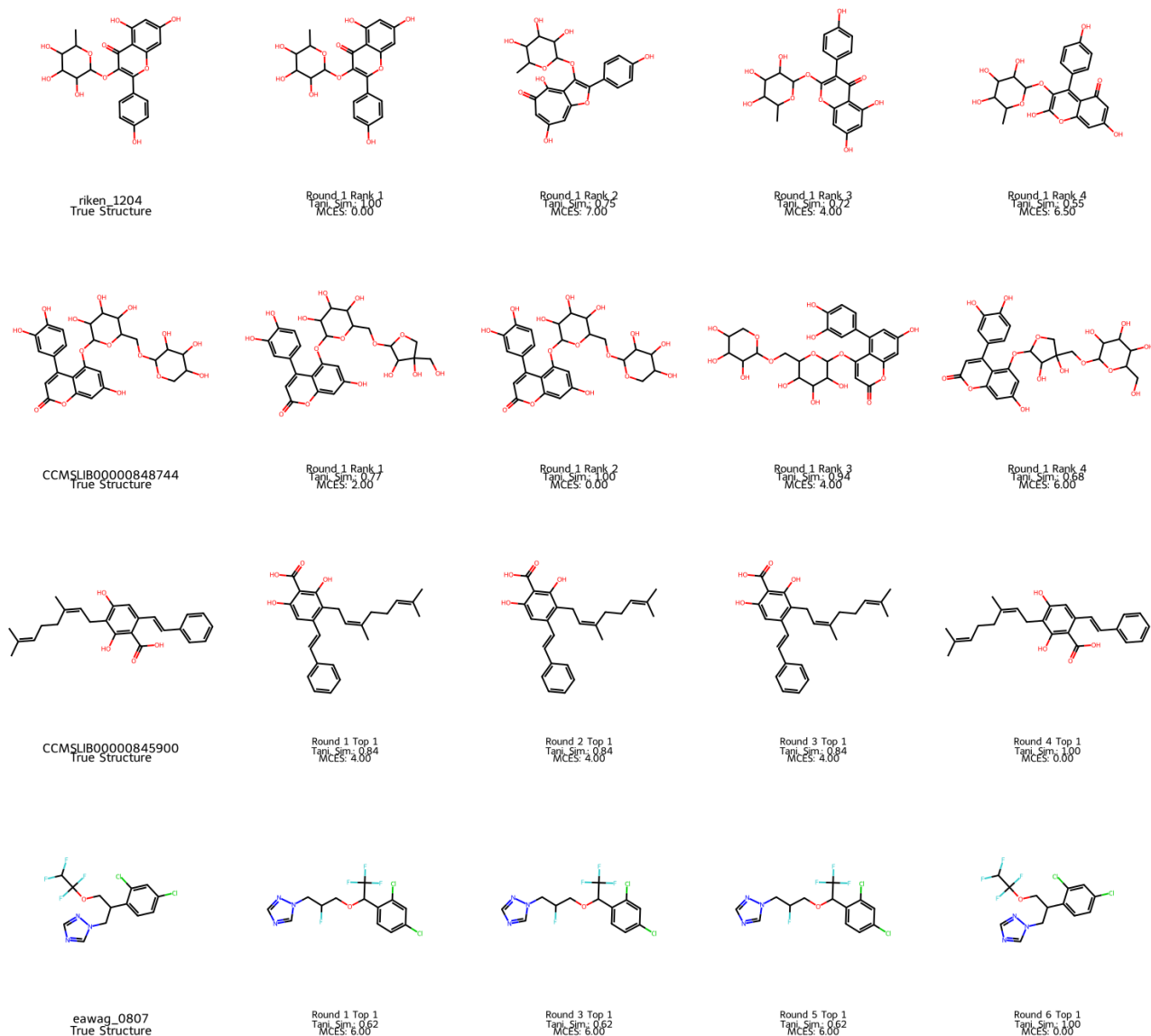


Figure 5. Examples of selected test spectra from the NPLIB1 dataset where FRIGID makes **successful** reconstructions. Rows #1 and #2: Two examples of immediate success, where the true structure was correctly generated and identified as the Top 1 (as in Row #1) or Top 2 (as in Row #2) candidate in Round 1 (where no inference-time correction has yet been applied). The panels here show the top 5 ranked candidates. Rows #3 and #4: Examples of an eventual success, where FRIGID fails to create the true molecule in earlier rounds, but eventually corrects its predictions to create the true molecule. Row #3 shows this eventual success after 3 rounds of inference-time correction; Row #4 after 5 rounds of correction.

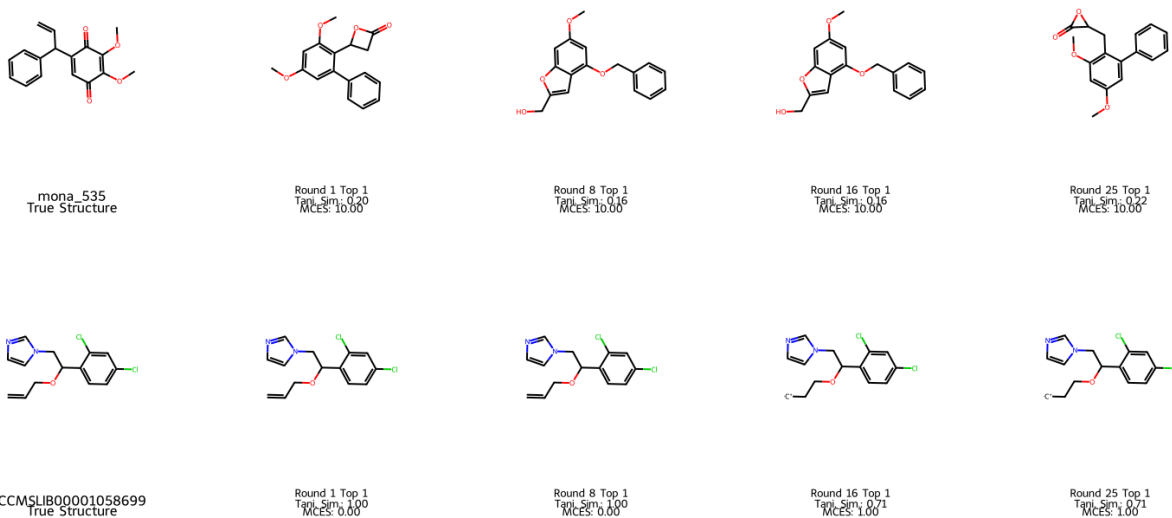


Figure 6. Examples of selected test spectra from the NPLIB1 dataset where FRIGID makes **errors**. Row #1: Failure example where FRIGID fails to make or rank the true molecule highly, even after 24 rounds of correction. Row #2: Failure case where applying inference time correction worsens the Top 1 ranked structure, from a true structure to a slightly mimicked decoy structure. Note that this structure technically differs in formula and would thus be excluded in a formula filtering step, but we visualize it to highlight a possible failure mode.

C.2. MassSpecGym Predictions

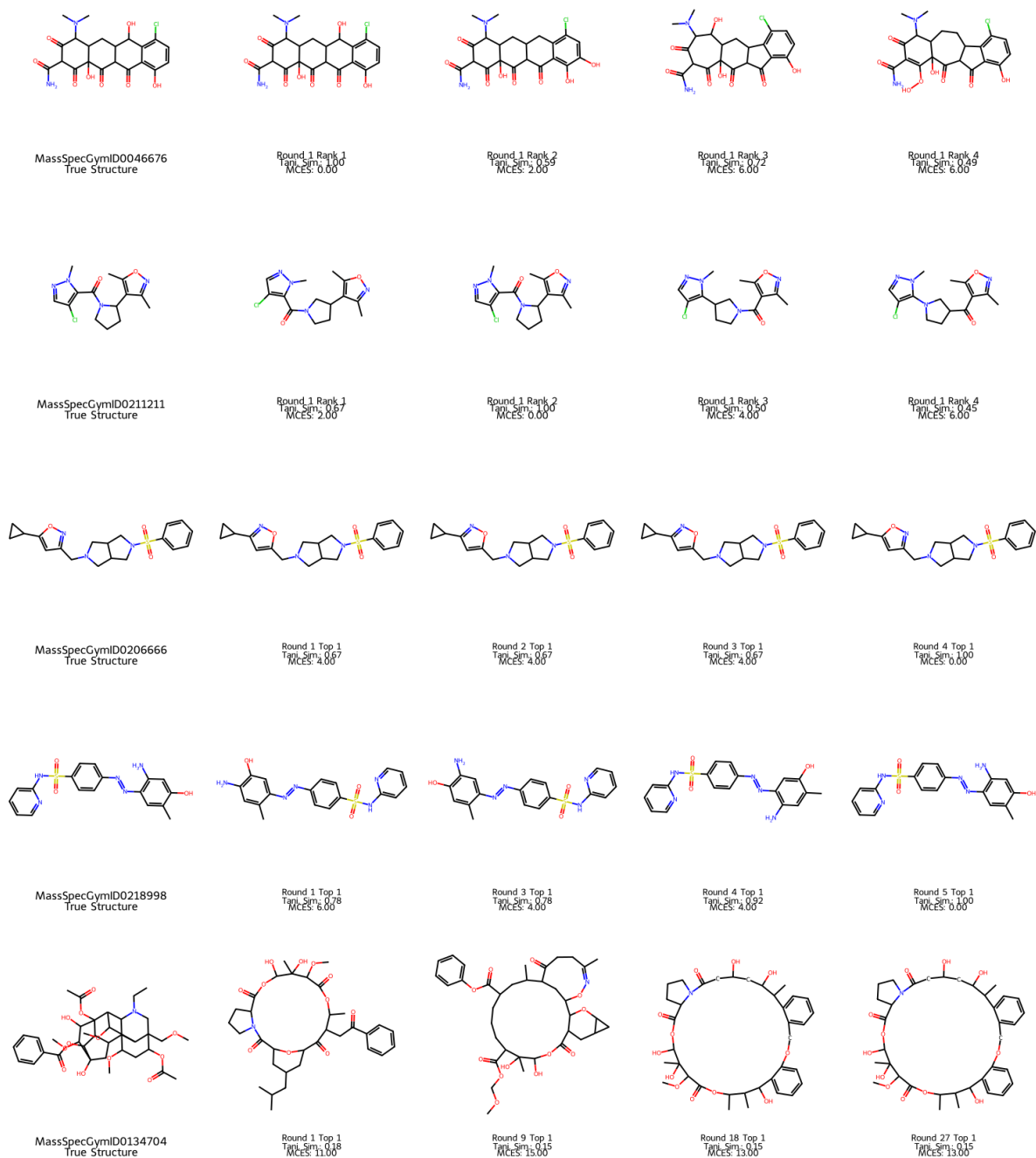


Figure 7. Examples of selected test spectra from the MassSpecGym dataset. Rows #1 and #2: Two examples of immediate success, where the true structure was correctly generated and identified as the Top 1 (as in Row #1) or Top 2 (as in Row #2) candidate in Round 1 (where no inference-time correction has yet been applied). The panels here show the top 5 ranked candidates. Rows #3 and #4: Examples of an eventual success, where FRIGID fails to create the true molecule in earlier rounds, but eventually corrects its predictions to create the true molecule. Row #3 shows this eventual success after 3 rounds of inference-time correction; Row #4 after 4 rounds of correction. Row #4's intermediate molecules show a progressive improvement in structural similarity leading up to the discovery of the true structure. Row #5: Failure example where FRIGID fails to make or rank the true molecule highly, even after 26 rounds of correction.